

# Presentation of



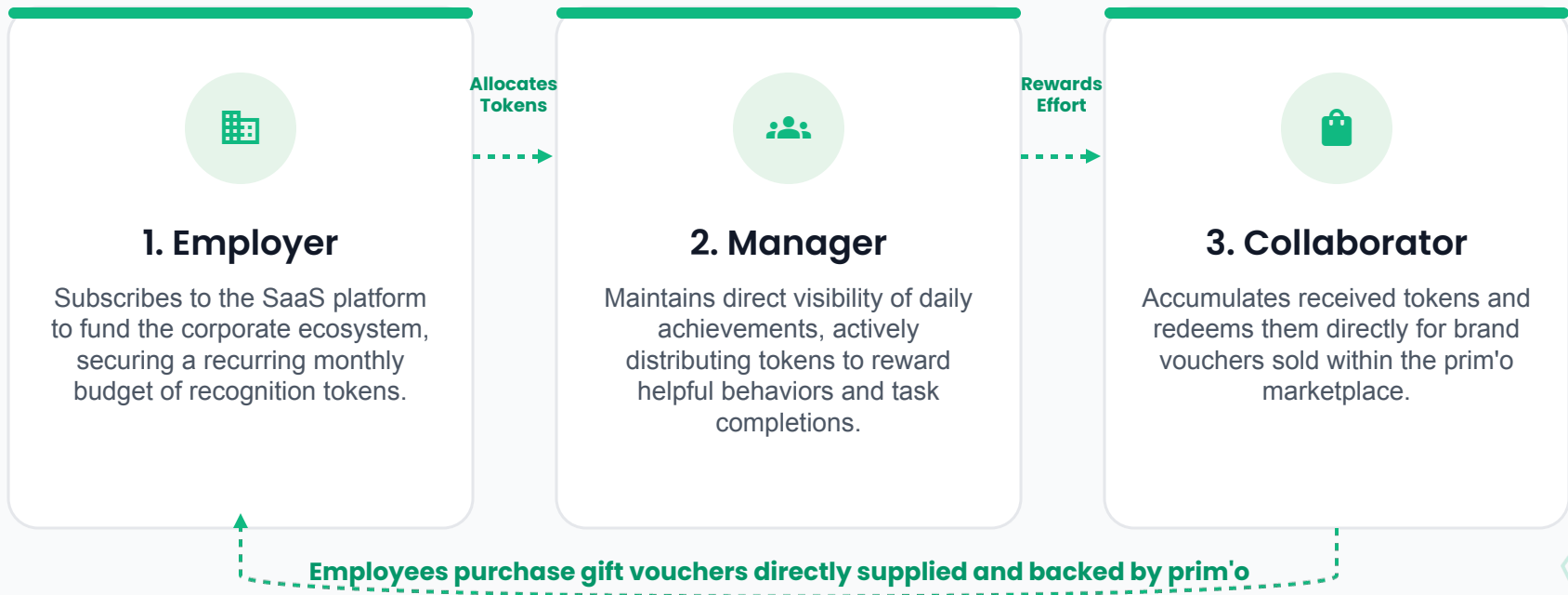
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Innovative reward solution bridging employee performance  
and instant recognition through tokenization



**Holberton**

# How **prim'o** Powers Value Distribution



OUR TEAM

# The Talents Behind prim'o



**Loïc Cerqueira**

Tech Lead

- Fullstack
- Guarantor of technical architecture



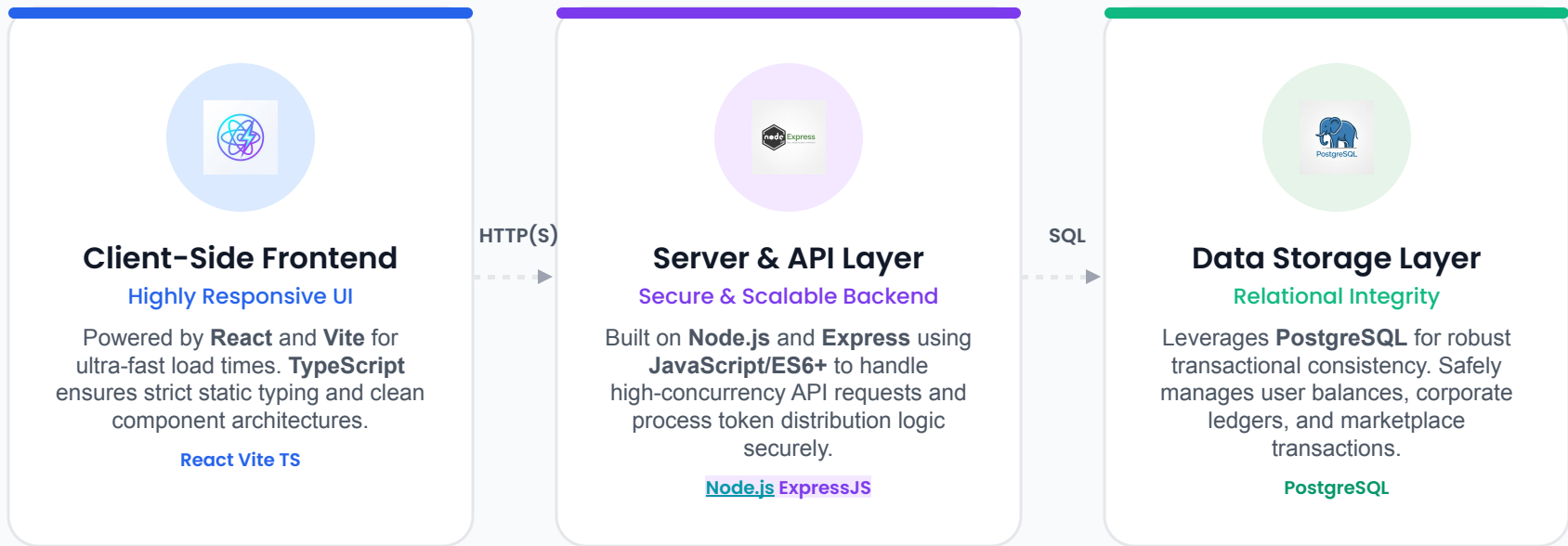
**Véronique Beauvais**

Fullstack Developer

- Fullstack
- Data design

## TECHNICAL ARCHITECTURE

# The Robust Tech Stack Behind **prim'o**



## ROADMAP & DEVELOPMENT PROCESS

# Our 4-Sprint Delivery Plan



Product Walkthrough

# MVP Demo



# request



## 1 routes/auth.routes.js

```
router.post(
  '/login',
  [
    body('email').isEmail()
      .withMessage('Valid email is required'),
    body('password').notEmpty()
      .withMessage('Password is required'),
    validate,
  ],
  authController.login
);
```

## 3 controllers/authController.js

```
const login = async (req, res, next) => {
  try {
    const data = await authService
      .login(req.body);
    res.json({ success: true, data });
  } catch (err) {
    next(err);
  }
};
```

## 2 middleware/validate.js

```
const { validationResult } =
  require('express-validator');

const validate = (req, res, next) => {
  const errors = validationResult(req);
  if (!errors.isEmpty()) {
    return res.status(400).json({
      error: errors.array()[0].msg,
      code: 400,
    });
  }
  next();
};

module.exports = { validate };
```

## 4 services/authService.js

```
const login = async ({ email, password }) =>
  const { User, Company, Team } =
    require('../models');
  const user = await User.findOne({
    where: { email: email.toLowerCase() },
    include: [{ model: Company, as: 'company'
  }]);
```

# Automatic Allocation : The Recurrence Engine

Code snippet -ScheduledAllocation.js

```
const initScheduledAllocation = (sequelize) => {
  ScheduledAllocation.init(
    {
      target_type: {
        type: DataTypes.ENUM('user', 'all_company', 'team', /* ... */),
        allowNull: false, // Target type (specific user, team, or whole company)
      },
      amount: {
        type: DataTypes.DECIMAL(10, 2),
        allowNull: false, // Token amount to allocate
      },
      frequency: {
        type: DataTypes.ENUM('monthly', 'annual'),
        allowNull: false, // Execution frequency (monthly or annual)
      },
      day_of_month: {
        type: DataTypes.INTEGER,
        allowNull: false, // Execution day of the month (1-28)
      },
      next_run_at: {
        type: DataTypes.DATE,
        allowNull: false, // Calculated timestamp of the next execution
      },
      active: {
        type: DataTypes.BOOLEAN,
        defaultValue: true, // Whether the rule is active or paused
      },
    },
    {
      sequelize,
      tableName: 'scheduled_allocations', // Table name in PostgreSQL
    }
  );
};
```

cron.service.js-runScheduledAllocations()

```
async function runScheduledAllocations() {
  // 1. Fetch all active rules that are due for execution (next_run_at <= now)
  const dueRules = await ScheduledAllocation.findAll({
    where: {
      active: true,
      next_run_at: { [Op.lte]: new Date() },
    }
  });
  for (const rule of dueRules) {
    // 2. Resolve target employees and perform the token transfers
    const targets = await resolveTargets({ /* ... */ });
    for (const receiver of targets) {
      await executeTokenTransfer(rule.sender_id, receiver.id, rule.amount);
    }
    // 3. Compute the next run date based on frequency (monthly/annual)
    const nextRun = rule.frequency === 'monthly'
      ? nextMonthly(rule.day_of_month)
      : nextAnnual(rule.day_of_month, rule.month);
    // 4. Update the rule with the next execution date
    await rule.update({ next_run_at: nextRun });
  }
}
```



# Results and Lessons learned

## ✓ **DEAL WITH THE UNEXPECTED**

Adapting quickly to project changes and solving problems as they arise.

## ✓ **USE AN EXTERNAL API**

Integrating powerful third-party services to expand app capabilities.

## ✓ **A FUNCTIONAL WEB APP**

Delivering a fully working prototype built on solid development principles.

## ✓ **THE WEALTH OF IT DEVELOPMENT**

Gaining a deep understanding of software design and modern frameworks.

## ✓ **USER EXPERIENCE & INTERFACE**

Designing elegant, user-centric interfaces that make operations intuitive.

# Problem encountered

## ⚠️ TECHNICAL DIFFICULTIES

Several technical issues arose throughout development, requiring time to debug and resolve.

## ⚠️ TIMING AND DEADLINE

Managing strict project schedules, milestones, and delivering key components under tight time constraints.

## ⚠️ DEPLOYMENT CHALLENGES

Handling deployment was a major challenge during the project.

## ⚠️ CONFIGURE AN EXTERNAL API

Overcoming authentication, rate limiting, and structure mismatches when integrating third-party endpoints.

# Future of the app

## ✓ **COOKIES**

Move JWT storage from localStorage to httpOnly cookies to reduce XSS risks..

## ✓ **REAL TIME NOTIFICATION**

Replace the current setup with real push.

## ✓ **FULL i18n**

A language toggle already exists in the preferences, but no i18n library is currently integrated to translate the interface.

## ✓ **USER RATING PAGE**

A dedicated rating page was created so users can score the app.

## ✓ **PERFORMANCE MONITORING**

Add a monitoring dashboard that aggregates key performance metrics.

PROJECT REVIEW

# The End

THANK YOU FOR LISTENING

Thanks to Julien SAX & Sandrine SIMONNET  
for this project

